Industrial Surface Defect Classification: Cross-Entropy and Poly-1 under Two Training Budgets

Course: AI Implementation Technologies

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Dataset: KolektorSDD2

Final experimental freeze: 12 November 2026

Study status: Original fixed-update study completed; independent 30-full-epoch follow-up completed for one random seed. No further training is included in this final scope.

Abstract

This study examines binary classification of industrial surface images (normal versus defective) using ImageNet-pretrained ResNet18 and two loss functions: standard Cross-Entropy (CE) and Poly-1. Two **distinct experimental protocols** are reported rather than merged. The initial fixed-compute study compares 70, 140, 280, and 420 training images at 216 optimizer updates, with three seeds and previously archived reference runs. After a concern about the limited number of complete dataset passes, an additional study trained all eight combinations of four image counts and two losses for **30 genuine full epochs** at seed 20261028. A common 180-image holdout drawn exclusively from the official training partition served for model selection in the follow-up. The new study achieved best internal validation Macro-F1 from 90.62–91.12% with 70 images up to 98.00% with 420 images. CE and Poly-1 tied at 280 and 420 images. These validation results are exploratory; neither demonstrates independent production performance.

1. Research question and scope

The objective is to compare CE and Poly-1 for normal/defective classification and investigate associations between training-set size and predictive metrics. The work is divided into two analyses with different compute budgets and evaluation populations. **Scores from these two analyses must not be compared as though they were computed on the same holdout.** The project is considered complete at the documented eight-run, single-seed follow-up; a larger 24-run extension is a possible future study, not completed work or an academic obligation established here.

2. Dataset and verification

KolektorSDD2 provides **2,331 official training images** (2,085 normal and 246 defective) and **1,004 official test images** (894 normal and 110 defective). The official source archive was checked against SHA256 edcdb486809b24f1d17b785e30c52fafc5999554dd5fe18ddf77b61ceb6f36a8. The selected training subsets contain 70, 140, 280, and 420 images, balanced equally between normal and defective and nested in ascending size. Exact image hashes were recorded for provenance; overlap by identical file content was excluded between selected training and the relevant evaluation subsets. **Physical-part or manufacturing-specimen independence was not established**, and equal hashes cannot detect all near-duplicates.

In the follow-up experiment, the same **180 validation images** were selected once from otherwise unused images in the official training partition, comprising 150 normal and 30 defective. This selection excludes the entire frozen 420-image training pool, so validation is disjoint from the 70-, 140-, 280-, and 420-image training subsets. The official test images were not used for follow-up model selection. One additional defective classification error changes defective recall by **3.33 percentage points**, making the reported recall sensitive to sample size.

3. Model and fixed settings

All experiments use **ResNet18**, initialized with torchvision ImageNet1K V1 weights, with a new linear two-class classification head and end-to-end fine-tuning. Input images are converted to RGB, bicubically resized to **448 × 448**, and normalized using ImageNet channel means and standard deviations. Optimizer: **AdamW**, learning rate 1e-4, weight decay 1e-4, batch size **4**. No data augmentation was used. Poly-1 follows the classification loss form **Poly-1 = CE + (1 - p_true)** with epsilon set to 1; p_true denotes the predicted probability assigned to the correct class. The follow-up holds the image order, seed, architecture and training hyperparameters constant within each CE/Poly-1 pair.

4. Study A — fixed optimizer-update budget

The initial study contained **24 evaluation records**, corresponding to four image-count conditions, two losses and three seeds. The 70-image source was an archived historical reference; **18 new checkpoint trainings** were produced for the other image-count conditions. Every condition used **216 optimizer updates** and 12 reference-equivalent rounds of 70 image exposures, not 12 complete passes through each larger training set. The effective complete-pass equivalents were **12, 6, 3, and 2** for sizes 70, 140, 280, and 420 respectively. BatchNorm running statistics were recalibrated under a fixed 70-image protocol. The values below are means across three seeds and relate to the **previously inspected** official 1,004-image test set.

Images	CE Macro-F1	Poly-1 Macro-F1	CE AUROC	Poly-1 AUROC
70	76.64%	75.72%	85.68%	85.62%
140	77.86%	79.24%	88.25%	88.95%
280	84.76%	86.10%	89.67%	90.80%
420	84.80%	86.07%	92.27%	93.07%

The strongest mean Macro-F1 in the fixed-216-update study was **86.10%** for 280 images with Poly-1, while the highest mean AUROC was **93.07%** for 420 images with Poly-1. Poly-1 had higher mean Macro-F1 at 140, 280 and 420 images but not at 70 images. The historical 70-image references were not retrained in the same numerical environment as the new checkpoints. These results describe the fixed-budget exploratory setup and do not establish causality or a blind held-out generalization result.

5. Study B — 30 complete dataset epochs

To address the limited effective passes in Study A, a **new, separate experiment** was conducted. Each model was trained for exactly **30 complete epochs** on every image in its selected training subset. For 70, 140, 280 and 420 images, this produces **540, 1,050, 2,100 and 3,150 optimizer updates per model**. Eight real full-epoch training runs were completed: four sizes × two losses × **one seed (20261028)**. The best checkpoint of each run was chosen at the earliest epoch achieving the highest **internal validation Macro-F1** across the 30 recorded epochs. Validation was measured on the **same 180-image set in all eight conditions**.

Images	Updates/run	CE best epoch	CE Validation Macro-F1	Poly-1 best epoch	Poly-1 Validation Macro-F1
70	540	23	91.12%	19	90.62%
140	1,050	12	91.29%	30	93.46%
280	2,100	7	95.77%	19	95.77%
420	3,150	30	98.00%	22	98.00%

At 420 images, both losses selected models with **98.00% validation Macro-F1** and **178/180 = 98.89% accuracy**. Their aggregate confusion matrices were each [[149, 1], [1, 29]] (ground truth rows: normal, defective). The 420-image CE model was selected from epoch 30; Poly-1 from epoch 22. The selected epoch by itself does not demonstrate general faster convergence. At 280 images the best Macro-F1 values were also equal (95.77%); Poly-1 was higher at 140 images (93.46% versus 91.29%), while CE was slightly higher at 70 images (91.12% versus 90.62%). **The follow-up does not show a consistent Poly-1 advantage.**

6. Interpretation

The second protocol directly resolves the concern that 420-image models had received only approximately two complete passes in the fixed-update study: both 420-image models were trained for 30 real full passes, and the other three sizes received the same nominal epoch count. The increasing internal validation Macro-F1 trend is consistent with useful additional training information, but this interpretation is limited in two major ways. First, the number of optimizer updates also rises with training-set size, so sample diversity and total update count are **confounded** in Study B. Second, each reported checkpoint was chosen using the same finite validation set; these are selected validation metrics rather than an independent final performance estimate. **The approximately 98% internal validation score must not be contrasted directly with the older 86% previously-used test score** because the protocols, data and selection rules differ.

7. Scientific limitations

- **Follow-up replication:** only one training seed is completed in Study B; there are no three-seed full-epoch averages or variability estimates.
- **Validation selection optimism:** the highest of 30 validation measurements was selected for each run.
- **Small defective validation sample:** 30 defective cases limit precision of recall comparisons.
- **Compute-size confounding:** 30 passes imply different total update budgets across training sizes.
- **Distribution shift:** balanced training sets do not reproduce the natural defect prevalence; the 150/30 validation mix is also a choice rather than a deployment estimate.
- **Specimen grouping:** physical-part independence and near-duplicate isolation are not fully verified.
- **Prior test exposure:** the 1,004-image official test had already been inspected in earlier experiments; Study A is exploratory and cannot be claimed as fresh blind confirmation.
- **Environment differences:** archived 70-image fixed-budget runs and subsequent conditions used historically different runtime environments.

8. Conclusion

A complete eight-run, 30-full-epoch follow-up was executed and saved separately from the original fixed-216-update study, without modifying historical outputs. The new data demonstrate that both losses can achieve **98.00% best internal validation Macro-F1** for the 420-image condition in this one-seed setup. No loss function consistently dominates across all four sizes. The original fixed-update study and the follow-up answer **different experimental questions**; both should be reported with their own evaluation population, compute budget and limitations. This work is final **for the agreed scope of eight full-epoch runs**, not for a hypothetical 24-run multi-seed extension. An untouched, specimen-separated evaluation set and multiple replicated seeds would be necessary for stronger generalization or statistical claims.

References

1. He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep Residual Learning for Image Recognition. CVPR. https://arxiv.org/abs/1512.03385 2. Leng, Z. et al. (2022). PolyLoss: A Polynomial Expansion Perspective of Classification Loss Functions. ICLR. https://openreview.net/forum?id=gSdSJoenupI 3. ViCoS Laboratory. Kolektor Surface-Defect Dataset 2 (KolektorSDD2). https://www.vicos.si/resources/kolektorsdd2/

Reproducibility and project locations

- Source code, protocols, archived results, audits: https://github.com/ID9999999999999/DL-Computer-vision/tree/nazih-training-set-size-20261109/COURSEWORK_NAZIH
- Original fixed-update checkpoints and records: Nazih_Coursework/Training_Size_Results, 01_Historical_70, 02_Evaluation_Results on Google Drive.
- New full-epoch checkpoints, training histories, frozen validation manifests and resume state: Nazih_Coursework/06_Full_Epoch_Pilot on Google Drive.
- New best-checkpoint SHA256 values are recorded in each COMPLETE.json. For the six newest full-epoch runs, remote content hashes have not been independently recomputed; do not claim a full remote bitwise audit.